PORTFOLIO

Mobile App Development Logbook

Weeks 1 — 7

June 2026

STUDENT SUBMISSION

Hadidja Aliani

BIT/2023/62116

Main project
Nikah Planner (Flutter)

Mini project
Student Attendance (SQLite)

Repository
github.com/moinaalia/nikah-planner

OVERVIEW

Executive Summary

This portfolio documents seven weeks of coursework in **Mobile Application Development**. The primary deliverable is **Nikah Planner** — a culturally appropriate wedding planning app built with Flutter, featuring authentication, multi-screen navigation, local persistence, and Firebase cloud integration. Weeks 6–7 extend the work with formal relational database design (Task 2) and a standalone **Student Attendance** application using SQLite (Task 3).

13	7	2
Screens built	Weeks logged	Projects delivered

Technology Stack	
Frontend	Flutter 3.x · Dart · Material Design

Navigation	go_router · bottom tabs · stack routes
Local data	SharedPreferences · SQLite (sqflite)
Cloud	Firebase Auth · Cloud Firestore
Version control	Git · GitHub (moinaalia/nikah-planner)

TIMELINE

7-Week Progress Map

1	2	3	4	5	6	7
W1	W2	W3	W4	W5	W6	W7

LOGBOOK

Weekly Activities

1 Week 1 — Requirements & UI Design

Scope · React prototype · User flows

WORK COMPLETED

- Wedding app scope (budget, guests, schedule)
- React UI prototype with gold/sage theme
- Screen flow: Splash → Auth → Dashboard
- CAT rubric mapping

KEY FILES

`src/` · README

DATE

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INSTRUCTOR MARK

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Comments: ____

2 Week 2 — Flutter Setup & Theme

SDK · Material · Reusable widgets

WORK COMPLETED

- Flutter project nikah_planner/

KEY FILES

- Theme: app_colors.dart, app_theme.dart
- Animated splash screen
- WeddingCard, PrimaryButton widgets

`lib/core/theme/` · `lib/features/splash/`

DATE

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INSTRUCTOR MARK

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Comments: ____

3 Week 3 — Auth & Navigation

Forms · go_router · Shell

WORK COMPLETED

- Login & register screens
- GoRouter named routes
- Auth flow and redirect guards
- Bottom navigation (5 tabs)

KEY FILES

`lib/features/auth/` · `lib/router/`

DATE

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INSTRUCTOR MARK

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Comments: ____

4 Week 4 — Core Screens

Lists · Charts · Filters

WORK COMPLETED

- Dashboard with countdown
- Budget pie chart (fl_chart)
- Guest list RSVP filters
- Schedule timeline · Settings

KEY FILES

`lib/features/home/` · `budget/` · `guests/`

DATE

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INSTRUCTOR MARK

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Comments: ____

5 Week 5 — Local Storage

SharedPreferences · JSON

WORK COMPLETED

- mock_data.dart demo profiles
- local_wedding_store.dart persistence
- WeddingRepository load/save
- RAM vs disk storage comparison

KEY FILES

`local_wedding_store.dart · wedding_repository.dart`

DATE

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INSTRUCTOR MARK

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Comments: ____

6 Week 6 — Firebase & DB Design

Auth · Firestore · Task 2

WORK COMPLETED

- Firebase Auth integration
- Cloud Firestore collections
- Offline/demo fallback mode
- Task 2: 4-table relational schema

KEY FILES

`auth_service.dart · firestore_service.dart`

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INSTRUCTOR MARK

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Comments: ____

7 Week 7 — SQLite Mini Project

sqflite · CRUD · Task 3

WORK COMPLETED

- student_attendance/ Flutter app
- Register students · mark attendance
- SQL JOIN attendance reports
- SHA-256 password hashing

KEY FILES

`student_attendance/lib/database/`

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INSTRUCTOR MARK

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Comments: ____

ASSESSMENT

CAT Rubric

Criterion	Max	Evidence	Mark
User Interface Design	5	13 themed screens	________
Navigation	4	GoRouter + 5 tabs	________
Event Handling	—	Forms, filters, gestures	________
Local Storage	5	SharedPreferences + SQLite	________
Data Retrieval	4	Repository + SQL JOINs	________
Networking / API	5	Firebase Auth + Firestore	________
Error Handling	3	AuthResult validation	________
TOTAL	**30+**		________

Instructor signature: ______________________________ **Date:** __________________

TASK 2

Database Design

Relational schema for a university student management system. Third normal form (3NF). Passwords stored as SHA-256 hashes.

students	student_id PK	full_name, year_of_study, phone, password_hash
courses	course_id PK	course_code UNIQUE, course_name
enrollments	enrollment_id PK	student_id FK, course_id FK
attendance	attendance_id PK	enrollment_id FK, date, status

Relationships: students 1→N enrollments N→1 courses · enrollments 1→N attendance

Task 2 mark: ____________ **Comments:** ______________________________

TASK 3

Student Attendance App

Stack: Flutter + sqflite + crypto

Features: Register students · Mark present/absent/late · Attendance % reports (SQL JOIN)

Security: Password hashing before SQLite INSERT

Register Student

Insert screenshot

student_attendance app

Mark Attendance

Insert screenshot

Date picker + status buttons

Attendance Report

Insert screenshot

Percentage table per student/course

Task 3 mark: ___________ **Comments:** ____________________________

EVIDENCE

Screenshot Gallery

Insert → Pictures → This Device

1. Splash Screen	2. Login
Insert image `01_splash.png`	**Insert image** `02_login.png`
3. Register	**4. Dashboard**
Insert image `03_register.png`	**Insert image** `04_dashboard.png`
5. Budget	**6. Guests**
Insert image	**Insert image**

05_budget.png

06_guests.png

7. Schedule

Insert image

07_schedule.png

8. Settings

Insert image

08_settings.png

9. Error State

Insert image

09_error.png

REVISION

Week 6–7 Questions

Q1. Mobile networking?

Client-server communication via HTTP/APIs (Firebase).

Q2. Client-server architecture?

Client requests; server processes; app never hits DB directly.

Q3. What is an API?

Rules for programs to exchange data.

Q4. What is JSON?

Text format: { "key": "value" }.

Q5. GET vs POST?

GET reads; POST sends/creates data.

Q6. SQLite?

Embedded relational DB on device with SQL.

Q7. Firebase?

Google BaaS: Auth, Firestore, Storage.

Q8. CRUD?

Create, Read, Update, Delete operations.

Q9. Local vs cloud storage?

Local = offline on device; cloud = internet sync.

Q10. Why databases in mobile?

Persistent structured data, queries, integrity.

Declaration

I confirm that this logbook represents my own work for the Mobile Application Development module, Weeks 1 to 7, including Nikah Planner and the Student Attendance mini project.

Hadidja Aliani · BIT/2023/62116

Signature: ___________________________ Date: ________________